

PORTFOLIO

CHARAN TEJA REDDY A M

May 19 2026

General Problems in Mathematics

1. Problem in a Plane mirror Field-of-view solved by using complex numbers.

Consider a plane mirror of fixed length a . Suppose a point source O is placed in front of the mirror as shown in fig 1.1 . It is also given that the ends of the

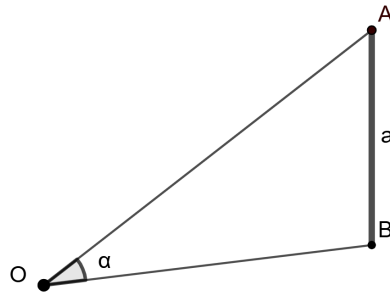


Figure 1.1: Geometric configuration of the mirror and the point source.

mirror make angle α at O when subtended to meet. There is a straight screen which is placed to exactly fit the field of view of O w.r.t the mirror. Furthermore, the ends of the screen when subtended to meet at the virtual image have equal lengths. Show that :

$$Screen\ length \propto |Sin(\frac{\alpha}{2})|$$